

Addendum: System Resonance Specifications (SRS) for Topological Resonance SLAM

Natasha Zink

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Overview

This addendum establishes the operational specifications for the Topological Resonance SLAM architecture. It defines the synchronization parameters between the physical hardware (OrchardOS) and the topological state space (F).

System Resonance Specifications

1. Harmonic Base Frequency (ν_0)

- **Target Frequency:** 18.08 Hz
- **Coherence Tolerance:** $\Delta\nu \leq 0.05$ Hz
- **Operational Purpose:** The system utilizes this frequency as the "Temporal Carrier" for all Fourier-Laplace operations. Deviations beyond the tolerance threshold trigger an automatic "Phase Correction" cycle to re-align the robot's internal hexagram state with the external environment.

2. Entropy-Energy Scaling Factor (Ω)

- **Definition:** $\Omega = \frac{\Delta E}{\mathcal{S}}$, where ΔE is the energy delta and $\mathcal{S}$ is the topological entropy.
- **Constraint:** The robot's movement is constrained by the Hofstadter fractal bands. If the entropy $\mathcal{S}$ of the sensor input exceeds the current resonance capacity, the system must apply a Topological Lift to move the path γ into a lower-entropy fractal band.

3. Feynman Interaction Amplitude (ψ)

- **Normalization:** $\int |\psi|^2 d\tau = 1$
- **Interpretation:** The amplitude ψ represents the interaction strength between the robot and localized environmental features.
- **Rule:** For any navigation vertex (change of direction or state), the robot must minimize $|\psi|^2$. This prevents the accumulation of "quantum drag" (unnecessary mechanical or algorithmic overhead).

4. Hexagram Transition Protocol (State Logic)

- **State Latency:** The system must resolve the state transition in the 64-hexagram cube within $\tau < 18.08$ ms of encountering a structural change in the environment.
- **Symmetry Constraint:** The transition must satisfy the Isocline Symmetry group requirements, ensuring the robot never enters a "forbidden energy state" (a gap in the butterfly manifold).

5. Resilience Protocol

- **"Bad Day" Logic:** In accordance with the Forest vs. Trees protocol, if the system's resonance operator R drops below the stability coefficient $\phi \approx 0.618$ (the Golden Ratio constant), the robot is required to enter a "Ground State" (stationary resonance) to perform a full Fourier-Laplace recalibration.

Algorithm: Topological Resonance Lift

The core navigation optimization is defined as:

$$R = \mathcal{F}_L(\psi_{\text{Feynman}})$$

$$\text{Minimize: } \int_{\gamma} R(\gamma) dt \rightarrow \min$$

where $\mathcal{F}_L$ represents the Fourier-Laplace transform, γ is the trajectory, and the optimization adheres to the fractal bands of the Hofstadter manifold.